

Python Code

Surface Blast Design Code

```
import math

# =====

# INPUTS

# =====

BD = float(input("Enter the Blasthole Diameter (mm): "))
d = float(input("Enter the Explosive Density (g/cm3): "))
rA = float(input("Enter the Spacing-to-Burden Ratio: "))
BH = float(input("Enter the Bench Height (m): "))
PFtech = float(input("Enter the Technical Powder Factor (kg/m3): "))
PPV = float(input("Enter the PPV (mm/s): "))
sRockType = input("Is the rock Hard or Soft? ").strip().lower()

# =====

# VALIDATION

# =====

if PPV <= 0 or PFtech <= 0 or rA <= 0 or BH <= 0:
    print("Error: All inputs must be greater than zero.")
    exit()

# =====

# CALCULATIONS

# =====

# Linear Mass Density (kg/m)
M = (d * (BD ** 2)) / 1273
```

Safe distance (m)

$D = ((700 / PPV) ** 1.67) * \text{math.sqrt}(M)$

Stemming Length (convert mm to m)

if $D < 1000$:

$SL = (30 * BD) / 1000$

else:

$SL = (20 * BD) / 1000$

Check that stemming is less than bench height

if $SL \geq BH$:

$\text{print}(\text{"Error: Stemming length is greater than or equal to the bench height."})$

$\text{exit}()$

Burden (m)

$\text{inside} = (M * (BH - SL)) / (rA * PF_{\text{tech}} * BH)$

if $\text{inside} < 0$:

$\text{print}(\text{"Error: Cannot calculate burden because the value inside the square root is negative."})$

$\text{exit}()$

$B = \text{math.sqrt}(\text{inside})$

Spacing (m)

$S = rA * B$

Subdrill (m)

$SD = 0.25 * B$

Actual Powder Factor

```

if sRockType == "hard":
    PFact = (M * (BH - SL + SD)) / (B * S * BH)
elif sRockType == "soft":
    PFact = (M * (BH - SL - SD)) / (B * S * BH)
else:
    print("Invalid rock type. Please enter Hard or Soft.")
    exit()

```

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# =====

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# OUTPUTS

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# =====

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print("\n===== RESULTS =====")
print("Linear Mass Density (M): {M:.2f} kg/m")
print("Safe Distance (D): {D:.2f} m")
print("Stemming Length (SL): {SL:.2f} m")
print("Burden (B): " + str(B) )
print("Spacing (S): {S:.2f} m")
print("Subdrill (SD): {SD:.2f} m")
print("Actual Powder Factor: " + str(PFact) + "kg/m3")

```

Fragmentation Calculation

Inputs

```

rA = float(input("Enter the rock factor: "))
iREE = int(input("Enter the relative explosive energy here: "))
PFtech = float(input("Enter the technical powder factor: "))
rP = float(input("Enter the explosive density here: "))

```

```
rK = float(input("Enter the K constant: "))
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```
rD = float(input("Enter the blasthole diameter (mm): "))
```

```
BH = float(input("Enter the bench height (m): "))
```

```
SL = float(input("Enter the stemming length (m): "))
```

```
# Calculations
```

```
M = (rP * rD**2) / 1273
```

```
Q = M * (BH - SL)
```

```
MeanFragmentationSize = (
```

```
    rA
```

```
    * (rK ** (-0.8))
```

```
    * (Q ** 0.167)
```

```
    * ((115 / iREE) ** 0.633)
```

```
)
```

```
# Output
```

```
print()
```

```
print("Mean Fragmentation Size =", round(MeanFragmentationSize, 3))
```